Interpretable Multi-Class ECG Arrhythmia Classification Using 1D CNNs and Grad-CAM

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***Abstract*—Cardiac arrhythmias require timely and precise diagnosis, yet deep learning models used for this task give no indication which part of a beat drove their prediction, leaving clinicians unable to validate outputs against their own diagnostic judgement. This paper investigates whether Grad-CAM can produce clinically meaningful explanations for a 1D Convolutional Neural Network on multi-class ECG arrhythmia classification. Existing high-accuracy approaches report limited interpretability analysis and rely on beat-level splits that risk patient-specific data leakage; this work instead uses deChazel's strict inter-patient split (DS1/DS2), yielding a lower but leak-free test accuracy of 77.64% on the Massachusetts Institute of Technology-Beth Israel Hospital (MIT-BIH) database across the Association for the Advancement of Medical Instrumentation (AAMI) classes. A manual implementation of Grad-CAM is applied to classified beats with activations interpreted against known ECG morphology and is validated through region-activation scoring and a faithfulness test. Results show class-specific activation patterns consistent with clinical hypotheses for V and F classes (faithfulness score up to 0.983), while S-class misclassification is linked to insufficient P-wave engagement and N-class produces no interpretable activation at all, a limitation of region-based explanation methods. A further finding was that faithfulness scores shift substantially with the placement of the Grad-CAM hook shows that Grad-CAM outputs can not be trusted unconditionally and must be interrogated rather than taken at face value.**

***Index Terms*—Interpretability, Arrhythmia, Inter-Patient Split, ECG classification**

I. Introduction

Cardiovascular disease is the leading cause of mortality in the UK, accounting for approximately 170,000 deaths annually (BHF, 2026). The most significant identifiers linked to this statistic are cardiac arrhythmias, abnormalities in the electrical conduction of the heartbeat that if undetected can lead to stroke, heart failure or sudden cardiac death. The Massachusetts Institute of Technology-Beth Israel Hospital (MIT-BIH) arrhythmia database annotates 15 unique arrhythmia classes, standardised by the Association for the Advancement of Medical Instrumentation (AAMI) which are reduced into five clinically meaningful beat categories: Normal (N), Supraventricular (S), Ventricular (V), Fusion (F), and Unknown (Q) (deChazal, O'Dwyer and Reilly, 2004). Electrocardiograms (ECGs) are the primary tool for detecting these arrhythmias, though interpreting these signals demands clinical expertise and is time-consuming.

Deep learning has made significant progress in automating ECG classification, where recent approaches are achieving high accuracies on datasets. However, predictive performance alone is largely insufficient for clinical deployment as internal decision-making of neural networks remains opaque to clinicians who may act on a model's output. In high-stakes healthcare environments, this opacity can have serious consequences: doctors cannot trace back through the model's reasoning, nor identify where it may have gone wrong. Additionally, the absence of interpretability prevents specialists from cross-checking the output against their own clinical judgement, a practice that is routine in traditional diagnostic medicine. Without this insight, no justified decision can be made based on a model's prediction alone, however high the reported accuracy. This work addresses ECG classification using a 1D Convolutional Neural Network (CNN) because it operates directly on the raw time-series data, preserving the temporal structure of the P-QRS-T waves that a 2D representation would otherwise need reformatting via signal transformation.

Explainable Artificial Intelligence (XAI) is one way that addresses this barrier. Several methods including Local Interpretable Model-agnostic Explanations (LIME), SHapley Additive exPlanations (SHAP), saliency maps and Grad-CAM are widely considered, but unlike perturbation-based or raw-gradient approaches, Grad-CAM generates class-discriminative visualisations directly from the learned convolutional features rather than treating it as a black box, making it a method adopted in this work. It produces visualisations of model decisions and could improve clinicians' trust in AI-based diagnosis as they can analyse and interpret how the model reached its decision. Therefore, in this paper we aim to determine: *To what extent can Grad-CAM produce medically meaningful explanations of a 1D Convolutional Neural Network (CNN) in multi-class ECG arrhythmia classification?*
The main objectives of this paper are as follows:

- Build a 1D CNN for five-class AAMI arrhythmia classification
- Apply class-discriminative Grad-CAM, adapted for 1D signals, across all five arrhythmia classes
- Evaluation of learnt representations against ECG morphology through qualitative and quantitative Grad-CAM analysis
- Benchmarking against state-of-the-art approaches

The rest of this paper is structured as follows: Section II will review related work in ECG classification and model interpretability. Section III describes the chosen methodology. Section IV presents results, Section V discusses them, and Section VI concludes the paper.

II. Related Work

A. Deep Learning for ECG Arrhythmia Classification

Prior to the adoption of deep learning ECG classification relied on manual feature extraction fed into classifiers such as support vector machines (SVMs) or random forests. Unlike these approaches, CNNs learn discriminative features directly from raw signal data, removing the need for feature engineering. Though 2D CNNs have been applied to ECG signals (represented through spectrograms or images), 1D CNNs operate directly on the raw signal, preserving temporal resolution with reduced computational overhead. Tang et al. establishes that kernel size is a critical parameter choice that governs a 1D-CNN's ability to capture temporal patterns at the appropriate scale for time-series classification (Tang et al., 2020). In the ECG domain, smaller kernel sizes have been shown to support extraction of the short-range temporal dependencies needed to capture P-QRS-T morphology (Murat Duranay et al., 2026).

Khan et al. (2023) applied a 1D residual neural network to the MIT-BIH Arrhythmia Database across the five AAMI mappings, using Synthetic Minority Over-sampling Technique (SMOTE) to address the severe class imbalance in the dataset. Though test accuracies are high, the results warrant scrutiny. A standard classifier outputs only a class prediction, with no built-in mechanism determining which input drove what decision. No saliency mapping or feature attribution was applied, leaving no indication of what P-QRS-T region informed its prediction. Thus, there are no means of determining whether the model reflects clinically meaningful ECG features or exploits spurious correlations. Also, neither the choice of a 1D CNN architecture nor SMOTE's class balancing offers any way to inspect what the model bases its prediction on, a critical omission needed for building any healthcare system. Furthermore, Khan et al. does not use an inter-patient split, the standard approach widely used in biomedical signal classification, where training and test sets are drawn from entirely separate patients to prevent data leakage (deChazal, O'Dwyer and Reilly, 2004). Instead, beats are distributed via stratified random sampling (Khan et al., 2023), meaning individual heartbeats from the same patient can appear in both training and test folds. Consequently, beats that appear in both sets allow the model to learn patient-specific patterns that optimistically produce biased evaluations of its diagnostic performance.

Subsequent work has compounded this problem through complex hybrid architectures. Kim et al. (2025) proposed a CNN-Transformer model using the Stockwell transform for frequency-domain feature extraction. Notably, their reported methodology does not specify whether patient-independent splitting was used. Similarly, Liu (2025) introduced ECG-Bert, another transformer-based model using bidirectional contextual encodings, to model long-range dependencies between heartbeat segments, dividing the dataset into training, validation and test sets, also without reported patient-level separation. Both echo the transparency gap leaving reported accuracies unreliable. Whilst this model outperforms across most metrics, it achieves the lowest V-class F1-score of all five classes. V-class represents ventricular ectopic beats characterised by a wide and bizarre QRS complex with a disassociated T-wave. Misidentifying this beat type risks a progression from ventricular tachycardia to fibrillation, a leading mechanism of sudden cardiac death (Samie and Jalife, 2001). The absence of interpretability analysis extends this failure because without it, there is no mechanism to understand why the most complex model fails on the most fatal class. A cardiologist cannot reason about the specific mechanisms that drive these decisions, and as architectural complexity increases, an inverse pattern emerges between performance and interpretability. Together these point to a broader issue in the field: accuracies have predominately become the primary optimisation target while neglecting transparency. High accuracy metrics do not constitute clinical validation and while works by Jahmunah et al. and Srinevasan and Charles (Jahmunah et al., 2022; Srinevasan and Charles, 2026) have begun applying post-hoc explainability directly to ECG-based classification, consistent absence of interpretability across this higher-accuracy benchmark represents the unaddressed gap that this paper seeks to address.

Table I summarises the architecture, evaluation split and interpretability of each benchmark alongside the proposed work.

TABLE I
Comparison of benchmark approaches and proposed work

Study	Arch.	Split	Acc.	XAI
Khan et al.	1D ResNet	Beat-level	98.63%	No
Kim et al.	CNN-Trans.	N/R	99.58%	No
Liu	1D-CNN+BERT	N/R	99.28%	No
This work	**1D-CNN**	**Inter-patient**	**77.64%**	**Yes**

B. Interpretability in Clinical AI

Clinician trust in diagnosis requires more than predictive accuracy; it also seeks transparency in how decisions are reached. Lipton argues that interpretability in machine learning is not a single defined property but a collection of distinct objectives, including "simulatability, decomposability, and algorithmic transparency" that all serve unique purposes (Lipton, 2018). Lipton identifies an evaluation problem within post-hoc interpretability specifically in that the literature largely avoids defining a *correct* explanation, often substituting subject plausibility for verifiable accuracy (Lipton, 2018). Identified nearly a decade ago this directly motivates the need for interpretability methods so outputs can be validated against ground truths, precisely the confirmation needed for identifying arrhythmia types on an ECG. Lipton (2018) argues that

in high-stakes environments interpretability is paramount for responsible deployment, reinforcing the clinical motivation of this work, where a model's prediction is insufficient without understanding how it was reached.

Decision trees are intrinsically interpretable models by design but at high cost to predictive performance, a documented interpretability-accuracy trade-off (Ennab and Mcheick, 2024). ECG arrhythmia classification is a complex task involving miniscule morphological changes between beats, making this trade-off costly and likely to exceed what intrinsically interpretable models can provide. Post-hoc explanation methods address this trade-off by applying interpretability analysis to an already trained model, preserving classification performance whilst providing a mechanism that interrogates learned representations.

Further support for post-hoc interpretations in the ECG domain is provided by Jahmunah et al. (2022), who applied Grad-CAM to a trained DenseNet model for myocardial infarction classification on the PTB database. Visualisations are mapped to specific ECG waveform component (P-wave, QRS complex, T-waves) that produce post-hoc explanations, so cardiologists can refer to and aid their decision making. Whilst Jahmunah et al. addresses a different clinical task, using a 12-lead ECG reading across ten myocardial infarction location classes, the underlying approach of applying Grad-CAM to CNN-extracted ECG features to named morphological components mirrors the methodology adopted in this work. Though prior work has applied Grad-CAM to arrhythmia classification, it remains less explored across all five AAMI classes under a strict inter-patient split. Murat Duranay et al. (2026) for instance applies Grad-CAM to arrhythmia detection directly, but only across four rhythm classes from the Chapman dataset rather than the more holistic five AAMI classes and inter-patient split used here.

C. XAI Methods for ECG Classification

Several post-hoc XAI solutions have been applied to interpret deep learning models, each with distinct characteristics that affect their suitability for 1D ECG classification, summarised in Table II. Though LIME and SHAP may seem useful, ECGs are time-series data that may contain several QRS complexes in a sample. Segmenting will disrupt this continuity and will misconstrue the original signal structure highlighting its inapplicable use in this context. Gradient-based methods offer a more natural fit for CNN architectures though not all are suited. Saliency maps and Guided Backpropagation operate at the raw input level rather than leveraging learned feature representations producing less meaningful results than Grad-CAM, which uses the final convolutional layer specifically to retain more spatial structure and encodes higher level, class specific features, generating class-discriminative heatmaps (Selvaraju et al., 2019).

Following this analysis, Grad-CAM is selected as the interpretability method for this work. A black-box method like SHAP and LIME only look at the model's inputs and outputs, never the internals, which would undermine the entire motivation for interpretability set out in this work. Srinevasan and Charles (2026) made a similar comparison between Grad-CAM and LIME on a 1D ECG classification model and found Grad-CAM effectively highlighted more clinically relevant temporal regions, including the QRS complex and ST-segment of ECGs - the exact morphological features needed for arrhythmia classification.

TABLE II
COMPARISON OF POST-HOC XAI METHODS FOR 1D ECG CLASSIFICATION

Method	Mechanism	Suitability
LIME/SHAP	Input perturbation	Low - disrupts signal continuity
Saliency/Guided Backprop	Raw input gradients	Moderate - lacks feature-level grounding
Grad-CAM	**Final-layer gradients**	**High - class-discriminative**

III. METHODOLOGY

A. Dataset and Inter-Patient Split

Out of 48 available in the MIT-BIH database, four records (102, 104, 107 and 217) are dropped because they contain paced rhythm that are not representative of natural arrhythmia morphology (deChazal, O'Dwyer and Reilly, 2004). The remaining 43 records are divided using what is commonly termed an inter-patient split, where 22 records are allocated to DS1 (used for training and validation) and the remainder to DS2 (exclusively for testing), with no patient appearing in both sets. This addresses the data leakage problem seen in the methodologies of Khan et al. and Liu (Khan et al., 2023; Liu, 2025). Following this split, DS1 yields 50,995 labelled beats and DS2 48,117, totalling to 99,112 beats across the dataset. DS1 is then further split on a beat-level into training and validation using standard 80/20 stratified split.

B. Signal Preprocessing

A Butterworth bandpass filter (0.5-40Hz) is applied to each raw ECG signal prior to beat extraction. This range was selected following visual comparisons against narrower and looser passbands on a sample (see Appendix A). Zero-phase filtering `filtfilt` and a Butterworth design were chosen over alternatives such as Chebyshev for its maximal flat passband and used to avoid distorting the beats morphology that may reduce the model's classification and interpretability accuracy. The passband works by removing baseline wander below 0.5Hz from patient movement, and high-frequency noise above 40Hz from muscle activity and electrical interference preserving the P-wave, QRS complex and T-wave. This approach follows the principle used by Pan and Tompkins' widely used real-time QRS detection algorithm that also relies on bandpass filtering but for a different purpose (Pan and Tompkins, 1985). Their filter targets a much narrower 5-15Hz passband, built to maximise QRS detection rather than preserving the full beat, including suppressing the T-wave content to eliminate any false positives that may inflate

QRS detections. This risk is avoided entirely by using R-peak locations already provided by the databases annotations.

Each signal is then z-score normalised per record, subtracting its mean and dividing it by its standard deviation, removing variations in signal amplitude.

C. Beat Extraction and AAMI Mapping

For each R-peak location given by .atr annotations, a 360-sample window is extracted spanning 180 samples either side, totalling to one second of signal at the databases 360Hz sampling rate. R-peaks within 180 samples of the start or end of a recording are dropped as a complete window cannot be extracted without exceeding the signal boundary. Each extracted beat is then labelled according to MIT-BIH annotation symbols at its corresponding R-peak, using the AAMI mappings established by deChazal et al (deChazal, O'Dwyer and Reilly, 2004), who reduced 15 heartbeat symbols into five clinically defined classes (N, S, V, F, Q). Beats annotated with symbols falling outside this mapping such as non-beats are removed.

Table III shows the class distribution across DS1 and DS2. Note the severe class imbalance toward the N-class, and the scarcity of F & Q beats which motivate the class balancing approach described in Section III-D.

TABLE III
CLASS DIST. ACROSS DS1 AND DS2 FOLLOWING AAMI MAPPING

Class	DS1	DS2	Total
N	45,841	42,668	88,509
S	944	1,836	2,780
V	3,788	3,218	7,006
F	414	388	802
Q	8	7	15
Total	**50,995**	**48,117**	**99,112**

D. Handling the Class Imbalance

Of the 50,995 beats displayed in DS1, the 80/20 split described in Section III-A allocates 40,796 beats to training. Though as aforementioned, the subset is still imbalanced: 36,673 Normal beats compared to 755 Supraventricular, 3,031 Ventricular, 331 Fusion and only 6 in Unknown. Not addressing this before training would result in the model leaning into a bias predicting the majority class, producing a model that appears accurate overall but systematically failing in most cases.

To combat this, SMOTE is applied to the training set. For each minority class, SMOTE identifies its `k_neighbors` nearest beats of the same class and interpolates a new synthetic beat between them, requiring a default of five neighbours (at least 6 beats). The Q-class sums to only 6 beats leaving no margin, so we change `k_neighbors` to two, providing room against this constraint. This approach is consistent with related works mentioned where Khan et al. and Kim et al. (Khan et al., 2023; Kim et al., 2025) both uses SMOTE to tackle the class-imbalance problem in the MIT-BIH database. All five classes now are balanced to the majority class with this oversampling technique, with a total yield of 183,365 beats, 36,673 per class. SMOTE is only applied to the training set, validation and test sets remain untouched to show their true imbalanced distributions to reflect the model's real diagnostic behaviour.

E. CNN Architecture

The model in Figure 1 visualises the architecture of a two-block 1D-CNN where each convolutional block follows the same structure: a convolutional layer, batch normalisation, ReLU activation, max pooling and dropout.

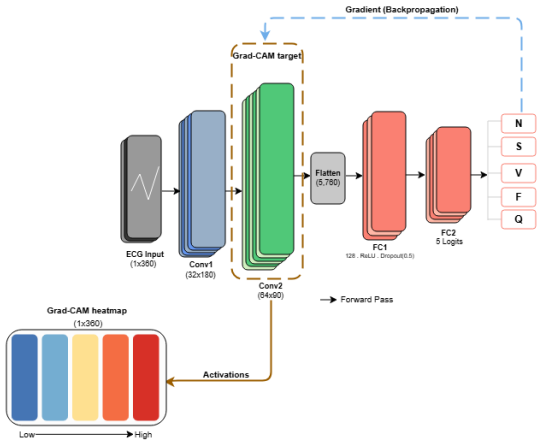


Fig. 1. CNN Architecture

`Conv1` takes a single-channel with a 360-sample input beat and expands it to 32 feature maps. Kernel size follows the rationale established in section II-A (Tang et al., 2020; Murat Duranay et al., 2026) with $k = 7$ selected to capture short-range P-QRS-T dependencies while remaining computationally efficient. Padding $p = 3$ is applied to each side of the signal following $p = \frac{k-1}{2}$ when kernel size $k = 7$, allowing the kernel to remain centred on every sample, outputting a total of $32 \times 360 = 11{,}520$ activations. `BatchNorm` standardises each feature map by approximating zero mean and unit variances. `ReLU` then introduces non-linearity by reducing all negative feature values to zero. `MaxPooling` with a kernel size of 2 then halves the signal length from 360 to 180 samples by retaining the strongest activation from each adjacent pairing, reducing the output to 32 feature maps $\times$ 180 (5,760 activations) simultaneously reducing computational cost. The regularisation technique `Dropout` is applied last that randomly deactivates half the neurons during training to prevent the network from relying heavily on specific neurons, reducing risks of overfitting.

`Conv2` repeats this structure but takes the relatively simple features extracted by Conv1 (a sharp local peak or a rising edge) and increases the feature map count from 32 to 64 to represent more detailed morphological structures (full QRS complexes or the transition into a T-wave) that Conv1's simpler features alone could not capture. Conv2s 64 filters spans across all 32 of Conv1's feature maps, with a kernel size of 32 input

channels × 7 temporal positions summing to 224 weights per filter. This allows Conv2 to learn combinations of patterns across Conv1s features dependently. `BatchNorm`, `ReLU` and `Dropout` are computed identically to the first block, followed by a second `MaxPooling` operation that halves the signal length again, reducing the output to 64 feature maps and 90 samples (5,760 activations).

The resulting feature maps and samples are flattened into a single vector of 64 × 90 = 5,760 activations. This vector is passed through a two-layer classification head consisting of two fully connected layers with non-linearity between them: `FC1` reduces the 5,760 inputs to 128 units, followed by ReLU activation and dropout. Finally, `FC2` maps the hidden layer to the final 5 output values per AAMI standardisation (N, S, V, F, Q).

The model was trained using the `Adam` optimiser (Kingma and Ba, 2014) to balance between convergence and training stability. Both network depth and its hyperparameters were not exhaustively tuned as this project's primary contribution is Grad-CAM interpretability. A simpler architecture keeps the relationship between learned features and Grad-CAM activations more directly traceable. `CrossEntropyLoss` was used, combining a softmax activation with negative log-likelihood loss that penalises weights in proportion to its confidence on an incorrect prediction. Table IV reports full set of hyperparameters for reproducibility.

TABLE IV
MODEL HYPERPARAMETERS

Hyperparameters	Value
Kernel Size	7
No. Blocks	2
Filters (Conv1/Conv2)	32 / 64
Optimiser	Adam
Learning Rate	0.001
Batch Size	64
Epochs	10
Dropout rate	0.5
Loss function	CrossEntropyLoss

F. Grad-CAM Implementation

Grad-CAM's implementation requires both forward and backward hooks directly on the `Conv2` layer, shown in the gradient path in Fig 1. The forward hook captures Conv2's activations during the forward pass while the backward hook extracts the gradients flowing into Conv2 during backpropagation. Together they provide the components needed to compute the heatmap. Conv2 is selected as the target layer following Selvaraju et al. (2019), who establish that the last convolutional layer offers the best balance of high-level semantics and detailed spatial information, unlike fully connected layers which flatten this. This mirrors the capacity discussed in Section III-E where Conv2's greater depth captures more morphological structures than Conv1 alone.

Grad-CAM was originally developed for 2D image classification, reflected in the design of `pytorch-grad-cam` (Gildenblat, 2021). Conv2's output in this model has shape (1, 64, 180) containing: a single batch, 64 feature maps and a temporal dimension of 180 samples. Attempting to apply the library directly to this input raises `ValueError: Invalid Input_tensor shape. Only 2D or 3D images are supported` thrown from the library's `get_target_width_height` which does not parse 1D inputs. This is resolved through the manual hook implementation described above which operates natively on Conv2's 1D output rather than requiring a 2D-oriented library.

Grad-CAM computes class-discriminative heatmaps by combining the gradients and activations from the hooks. The importance weight for feature map k is calculated by global-average-pooling its gradients over the temporal dimensions (Selvaraju et al., 2019):

$$\alpha_k^c = \frac{1}{Z} \sum_i \frac{\partial y^c}{\partial A_i^k} \tag{1}$$

where y^c is the raw pre-softmax score for the target class c, collected from the model's logits since softmax normalisation would dampen the class-discriminative gradient signal. Confidence values in results are then applied via softmax to these same logits. A_i^k is the activation at temporal position i within feature map k of the Conv2 output and $Z = 180$ is the number of temporal positions being averaged over.

These weights are then used to compute a weighted combination of the forward activation maps, followed by ReLU (Selvaraju et al., 2019):

$$L^c_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right) \tag{2}$$

where A^k attributes the full activation mapping of feature k, and ReLU retains only positive contributions of class c as single negative values are evidence of a different class (Selvaraju et al., 2019). The results $L^c_{\text{Grad-CAM}}$ map a 180-length heatmap, which is then upsampled to the original 360-sample beat length. Activations are captured immediately after the Conv2 operation before BatchNorm and ReLU, preserving samples of 180 rather than 90 that would remain after MaxPooling. This placement was chosen after an alternative post-normalisation hook was tested and found to degrade reliability. Though non-linearity is not introduced, the final ReLU in Eq.2 filters the result regardless.

To interpret the resulting heatmaps Grad-CAM produces, region boundaries for the P-wave, QRS complex and T-wave need to be derived from the standard P-QRS-T interval durations (Rawshani, 2017) and converted to samples using the 360Hz sampling rate where each sample corresponds to $1/360 \approx 2.78ms$. For example, a QRS duration of approximately 100ms (Rawshani, 2017) is positioned symmetrically around the R-peak at the windows centre (180):

$$\frac{100\text{ms}/2}{2.78\text{ms}} \approx 18 \text{ samples}, \quad 180 \pm 18 = [162, 198] \tag{3}$$

P-wave and T-wave boundaries are derived identically from their respective interval durations giving fixed ranges of 101-162 (P-wave) and 198-326 (T-wave). This allows quantitative region scores to be compared directly without introducing subjective, beat-by-beat boundary judgements that could steer bias towards stated hypotheses. However, these ranges only reflect normal sinus timing and may not correctly reflect the pathological morphology of arrhythmic beats. Consistent with the fixed-interval sampling approach used by deChazel (deChazal, O'Dwyer and Reilly, 2004) these boundaries are treated as approximate reference regions rather than precise per-beat ground truths.

Activation patterns are hypothesised prior to analysis based on ECG morphology, framing Grad-CAM as a test of clinical predictions rather than pattern-matching. V-class beats are expected to show activations concentrated around the T-wave and QRS complex. S-class beats are expected to show activations of the P-wave reflecting depolarisation produced in an ectopic atrial region. F-class beats are a blend of N and V so hypotheses reside within a mixed QRS and T-wave activation and finally, N-class activations are not expected to be interpretable given its dominance in the dataset.

A Grad-CAM heatmap cannot be the end point of analysis. Although activations can be seen, qualitative viewing alone is not enough to confirm the model is reasoning from meaningful ECG features. Three analyses therefore are performed using the trained model. Analysis 1 collects one correctly classified beat per class into a dictionary `found`, populated by iterating through DS2 predictions until each class is stored. Analysis 2 mirrors this process but collects three cases where a true S-class beat is misclassified as N. Analysis 3 quantifies these activations across all classes: for each beat in `found`, the proportion of total Grad-CAM energy falling within each region is calculated, giving a percentage score per region, per class.

Alongside Analysis 3, this work identifies a faithfulness test as a second metric to evaluate Grad-CAM's usefulness. While the former region-activation tests whether highlighted regions align with known ECG morphology, the latter determines whether those regions drove the model's prediction. Lipton points out that this is a wider problem in interpretability research where its rarely defined what a "correct" explanation truly means (Lipton, 2018). One way research addresses this problem is through a faithfulness score, a test of whether explanations truly reflect what the model is doing, not just whether it looks reasonable. Selvaraju et al. uses this method by masking small patches across an image and checking whether the patches change the model's score the most are the same ones Grad-CAM concentrated on (Selvaraju et al., 2019). This work adapts that idea directly to Grad-CAM's output. Ennab and Mcheick (2024) name this test selectivity: by removing a region deemed important should reduce the model's prediction confidence. For each beat in `found` (excluding Q for its statistical insignificance), the region above a 0.5 activation threshold is masked and the model's confidence in the true class is recorded before and after. As a drop in confidence does not prove Grad-CAM's success, so this drop is compared to an identical random region alongside region-activation scoring. If Grad-CAM's guided mask causes a bigger drop than a random one, it strongly confirms the highlighted region really is what the model is driving its prediction on.

IV. Results

A. Classification Performance

The trained model achieves a validation (DS1) accuracy of 99.23% compared to a test accuracy of 77.64%. A full per-class performance is reported in Table V. Beat-level splits risk letting the same patients appear in both training and test sets, allowing the model to memorise patient-specific patterns instead of morphology that generalises (deChazal, O'Dwyer and Reilly, 2004). DS1 and DS2 avoid this and share no patients. This gap reflects genuine generalisation to truly unseen physiology and highlights the inflated results that beat-level splits from Khan et al. and Kim et al. (Khan et al., 2023; Kim et al., 2025) produce and poorly reflects true diagnostic performance. The full training and validation curve across all epochs is provided in Appendix B.

TABLE V
DS2 CLASSIFICATION PERFORMANCE PER CLASS

Class	Precision	Recall	F1	Support
N	0.9465	0.7995	0.8668	42668
S	0.0394	0.1187	0.0592	1836
V	0.6318	0.9049	0.7441	3218
F	0.0609	0.2964	0.1011	388
Q	0.0000	0.0000	0.0000	7

Table V reports precision, recall and F1-score per class on DS2. V-class achieves the strongest recall (0.9049) amongst arrhythmia classes, while S-class collapses almost entirely, missed on 88.13% of occurrences (F1 = 0.0592). Q-class reports recall of 0.0000, reflecting its near total absence from the training data (Table III).

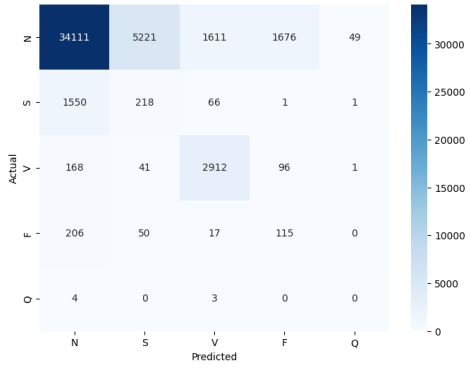


Fig. 2. DS2 Confusion Matrix

Figure 2 shows the DS2 confusion matrix. The most eye-opening error is the misclassification of S-class beats as N, accounting for 1550 of 1836 true S-class beats (84.4%). Section IV-C examines this error in more depth.

B. Grad-CAM Qualitative Analysis (Correct)

Q-class is excluded from this analysis, as no correctly classified Q beats exist in DS2 (Table V).

Figure 3 shows a correctly classified N-class beat. Activations remain at zero across the entire signal. N-class and V-class beats are both predicted with 100% confidence, yet V-class (Figure 5) produced strong structured activation, therefore confidence does not explain Ns flat result. A more likely explanation is that N-class beats do not have a single defining morphological feature for the model to attribute it to, unlike V-classes characteristically wide QRS complex, leaving Grad-CAM with no concentrated region to highlight.

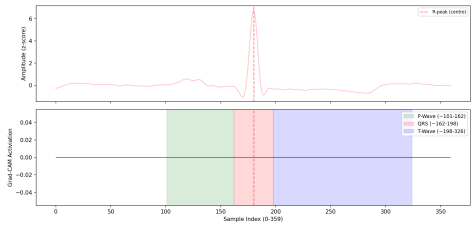


Fig. 3. Grad-CAM heatmap, correctly classified N-class beat

In Figure 4, the softmax confidence is computed at 96.57%. Activations concentrate sharply around the QRS complex with minimal engagement in the P-wave region. This contradicts the stated hypothesis and foreshadows the S-N failure mode examined in Section IV-C.

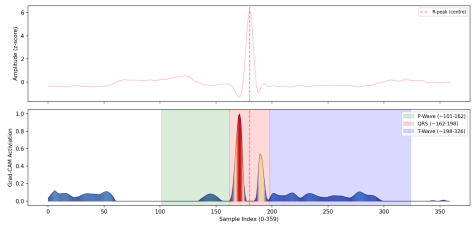


Fig. 4. Grad-CAM heatmap, correctly classified S-class beat

V-class (Figure 5) as aforementioned, has a predicted confidence of 100%. Grad-CAM activation concentrates mainly within the QRS complex with broad activation around the T-wave, consistent with the stated hypothesis of a ventricular ectopic beat.

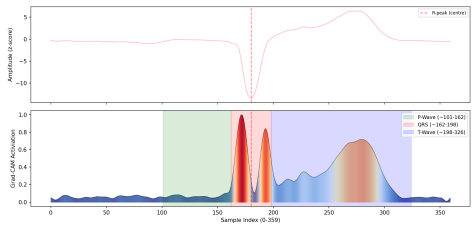


Fig. 5. Grad-CAM heatmap, correctly classified V-class beat

Finally, Figure 6 is predicted with 60.76% confidence, the lowest of N, S and V-class, highlighting a morphological blend as opposed to a distinct pattern. Activations appear moderate around the P-wave and sharper in the QRS and T-wave regions, in line with the mixed pattern of F-class reflecting a blend of N and V-class morphology.

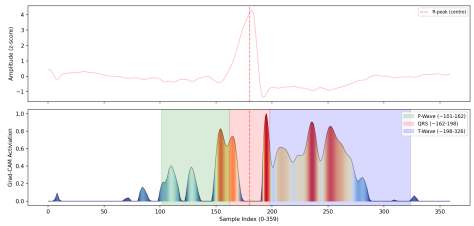


Fig. 6. Grad-CAM heatmap, correctly classified F-class beat

Three of four hypotheses are supported by Grad-CAM activations, though the S-class result reveals a discrepancy investigated in the following section.

C. Grad-CAM Qualitative Analysis (Failure Case)

Figure 7 shows an example of a true S beat misclassified as N. Activations concentrate on the QRS region with minimal P-wave engagement, the opposite of expected dominant P-wave activations in S-beat morphology. Two further misclassified examples shown in Appendix C show similar patterns.

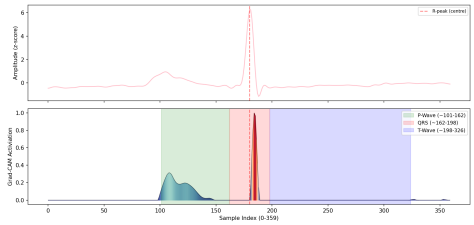


Fig. 7. Grad-CAM heatmap, S-class misclassified as N

D. Quantitative Region Scoring

TABLE VI
REGION-ACTIVATION ANALYSIS (% OF TOTAL GRAD-CAM ACTIVATIONS)

Region	N	S	V	F	S→N
P-wave	0.00	4.76	5.24	21.27	42.63
QRS	0.00	44.91	25.96	14.11	49.21
T-wave	0.00	28.85	59.54	62.31	4.23

Table VI above quantifies the qualitative heatmaps discussed above, reporting proportions of total Grad-CAM activations within each morphological region. V-class shows strong combined QRS and T-wave engagement, matching the expected wide QRS and oppositely directed T-wave pattern of ventricular ectopic beats. F-class shows the highest combined T-wave and QRS presence, also consistent with its hypothesis. S-class shows a heavy QRS focus with minimal P-wave engagement, reinforcing the original hypothesis contradiction noted above. However, the S-N failure cases show unexpectedly high P-wave figures nearly matching QRS. The cause of this reversal is taken up in Section V.

E. Faithfulness Evaluation

TABLE VII
FAITHFULNESS RESULTS PER CLASS

Class	GRD	RRD	Faithful
N	0.000	0.000	No
S	0.403	0.036	Yes
V	0.983	0.000	Yes
F	0.608	0.532	Yes

Results of the faithfulness test are reported in Table VII, comparing the confidence drop from masking Grad-CAM-guided region (GRD) against an equally masked random region (RRD). V-class shows strongest performance with GRD causing a larger drop than RRD. S-class also passes despite qualitative concentrations on QRS rather than the predicted P-wave, though clinically unexpected, it still meaningfully drove the model's prediction. F-class also passes by a smaller margin suggesting the model's decision may have spread broader across the beat rather than in one region. Finally, N-class could not be tested as no region exceeded the activation threshold.

V. DISCUSSION

The results suggest that Grad-CAM's medical usefulness is not uniform across all arrhythmia classes: while producing clinically clear explanations where classification succeeded (V, F) it diminished value for classes where either no single feature existed (N) or the model's true reasoning diverged from clinical expectation (S).

Despite the 100% prediction confidence of the N-class, Grad-CAM produced zero activation across the entire beat. This is not a failure of methods but a structural limitation of region-based interpretability, Grad-CAM highlights presences of a distinguishing feature, and N-class beats are defined by the absence of one. For the most common class a clinician would encounter in practice, Grad-CAM offers the least to work with, precisely where volume would be highest.

The faithfulness results (Table VII) support this picture. Three of the four testable classes showed faithful explanations, meaning the highlighted regions did not only look plausible but drove the model's confidence, while N remained untestable due to its lack of activation.

Furthermore, $S \rightarrow N$ failure mode is unlikely to stem from a single cause. DS1 contains only 944 real S-class beats in total (Table III), limiting how well the model can generalise P-wave morphology before facing unseen patients. The underlying class imbalance expands this since with only 755 real S-class training beats SMOTE can only interpolate within that limited pool meaning the augmented data may still miss genuine inter-patient P-wave variation. The confusion matrix (Fig 2) supports this: 84.4% of S-class beats misclassified as N specifically suggesting the model defaults to QRS-dominant boundary when P-wave evidence is weak. This holds even outside failure cases, as correctly classified S-beats lean heavily on the QRS complex in the region-activation results in table (VI) implying the model may never have relied on true S-class morphology at all. Yet the $S \rightarrow N$ failures, P-wave activation is substantial and the model still defaults to N. P-wave evidence therefore might be failing to anchor either way weak or present pointing to a limitation in the fixed region boundary, since these are derived from normal sinus timing and might not align with P-waves true position on ectopic beats.

Also, record 231 was inadvertently omitted from DS2 during data preparation leaving a 21-record rather than the standard 22 record test set. Though its omission is unlikely to affect reported trends.

As aforementioned in Section III-F, hooking onto Conv2's output after BatchNorm and ReLU rather than before caused V-classes faithfulness score to collapse to zero, degrading the reliability of the test. Though both hooks operate on equivalent activations they yield noticeably different quantitative results echoing Lipton's observations on what constitutes "correct" (Lipton, 2018). Therefore it may suggest Grad-CAM's quantitative outputs can be sensitive to implementation-level choices such as exactly where activations are captured relative to normalisation and non-linearity in ways not visible from a single heatmap.

Altogether these findings support this work's central premise: post-hoc interpretability can validate whether a model's decisions are clinically grounded in a way that accuracy figures alone cannot. Where Khan et al., Kim et al. and Liu report accuracies exceeding 98% with no means of inspecting what drove those predictions (Khan et al., 2023; Kim et al., 2025; Liu, 2025) our proposed methodology's modest 77.64% is accompanied by a direct, testable evidence of what the model actually learned. This also means Grad-CAM cannot be trusted unconditionally: as the hook-sensitivity finding shows, its own output depends on implementation choices, not just what the model actually learned.

VI. CONCLUSION

This work set out to establish whether Grad-CAM outputs can serve as evidence for a 1D CNN under a strict inter-patient evaluation. The model achieved a genuine 77.64% test accuracy on unseen patients, and the accompanying Grad-CAM and faithfulness analyses show that post-hoc interpretability can validate and challenge what a model has learned. This contribution is best understood as a proof of concept for Grad-CAM as a diagnostic tool for model trustworthiness rather than a finished clinical solution. As ECG classification models push closer to perfect accuracy, the real question stops being whether they are right and starts being whether we can ever know why.

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APPENDIX A: BANDPASS FILTER COMPARISON

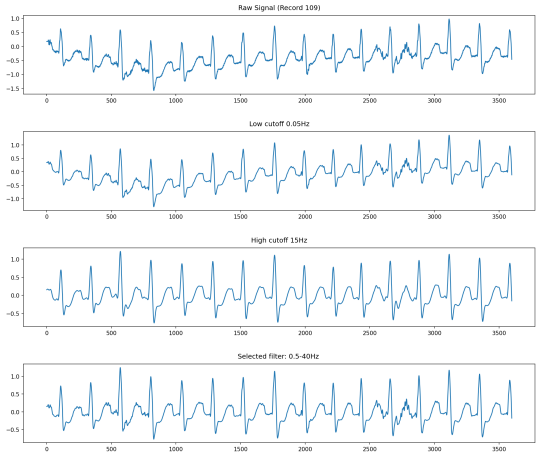


Fig. 8. Comparison of bandpass filter cutoffs on record 109: raw signal, loose cutoff, aggressive cutoff, and the selected 0.5-40Hz passband

This figure shows the visual comparison mentioned of varying bandpass filters evaluated in Section III-B. The selected passband removes baseline wander in the raw signal while preserving sharper morphological detail.

APPENDIX B: TRAINING AND VALIDATION PERFORMANCE

TABLE VIII
TRAINING AND VALIDATION PERFORMANCE PER EPOCH

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	0.1322	0.9562	0.0676	0.9822
2	0.0611	0.9804	0.0685	0.9817
3	0.0461	0.9850	0.0448	0.9882
4	0.0374	0.9878	0.0378	0.9914
5	0.0329	0.9894	0.0535	0.9885
6	0.0289	0.9906	0.0469	0.9902
7	0.0259	0.9917	0.0470	0.9895
8	0.0242	0.9921	0.0418	0.9906
9	0.0223	0.9930	0.0449	0.9914
10	0.0217	0.9929	0.0378	0.9923

Table VIII reports training and validation loss and accuracy across 10 epochs. Train loss decreases consistently while validation climbs to 99.23% accuracy by the final epoch matching reported DS1 accuracy reported in Section IV-A with no signs of overfitting.

APPENDIX C: ADDITIONAL S→N MISCLASSIFICATION EXAMPLES

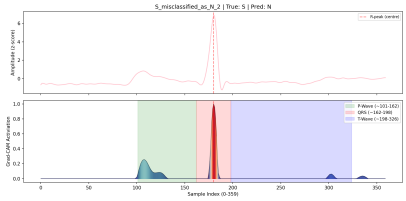


Fig. 9. Grad-CAM heatmap, second example of an S-class beat misclassified as N

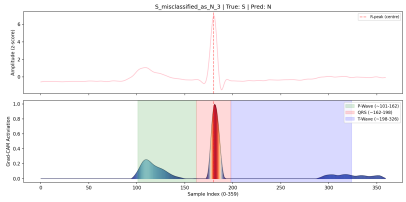


Fig. 10. Grad-CAM heatmap, third example of an S-class beat misclassified as N

Figures 9 and 10 correspond to the two misclassified examples referenced in Section IV-C. Both show the same QRS dominant, minimal P-wave engagement pattern in Figure 7.

APPENDIX D: DS1 VALIDATION CONFUSION MATRIX

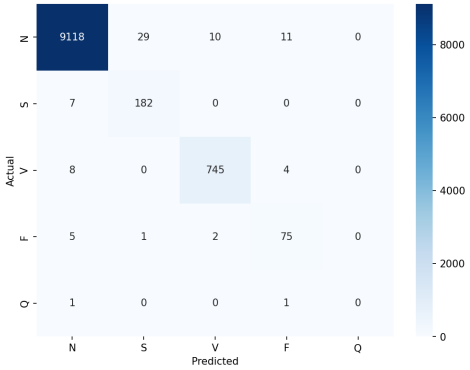


Fig. 11. DS1 validation matrix to compare against DS2 inter-patient split result

This confusion matrix aligns with the 99.23% accuracy reported in Section IV-A. It clearly shows how the generalisation gap between inter-patient split and beat-level splits visually compare from this work and Khan et al.

ECS750P/ECS753P/ECS754P– MSc Project Module
Generative AI Tools – Student Accountability Statement

The completed and signed statement must be included in the appendix of the final dissertation paper.

Student Name: Somaan Salim Mirza
Project Title: Interpretable Multi-Class ECG Arrhythmia Classification Using 1D CNNs and Grad-CAM
Supervisor Name: Dr. Somayyeh Timarchi

1. Have you used Generative AI tools for your project? YES
If yes, please complete sections 2 and 3 and sign the declaration below.
If no, please sign the declaration below.

2. Please provide details of all the AI tools that you used in your project work.

AI Tool	Version	Publisher	Access URL
Claude	Sonnet 4.6	Anthropic	https://claude.com

3. If you have used Artificial Intelligence (AI) tools in any part of your project please include in table format how, where, and why these tools were used. This should include references to the relevant parts of your project work — for example:

- Specific chapters or sections of the report (e.g. literature review, methodology, requirements)
- Software development (e.g. AI-assisted code generation, debugging, documentation)
- Interview or survey materials (e.g. AI-generated or refined questions/prompts)
- Datasets (e.g. AI-assisted data cleaning, labelling, or synthesis)
- The showcase video (e.g. AI-assisted scripting, narration, or editing support)

Be specific and transparent about your use of AI tools, including both direct outputs (e.g. text, code, images) and indirect uses (e.g. idea generation, editing assistance). The goal is to ensure academic integrity and provide a clear record of how AI contributed to your work.

Example Table – please add your own

Project Section	Description	Justification
Full Report	Grammar, punctuation	Used claude to catch recurring errors (comma splices, missing possessives) that may of disrupted the flow of writing though all were verified by myself before implementing.
Methodology (Grad-CAM implementation)	LaTeX formatting and equation rendering	Hadnt used LaTeX to write equations out especially in a two-column format, so it assisted me to debug the technical errors I

		was facing. Equations are my own or correctly cited as a reference.
Code debugging	Used ai to assist with the error of the Grad-CAM implementation where it required 2D input opposed to 1D	Merely used to help me debug as it was a critical error and other alternatives online did not solve the issue. All code related analysis were written myself.
LaTeX tables	Tables, Figures that were over the \columnwidth of the two-column page	Gave me a template to ensure they all looked the same
Presentation script	In my first iterations of the presentation I was exceeding the 10 minute limit including the +/- 10%	It took my already written script and gave me suggestions where I could cut certain sections out, but all decisions had been my own

3. Generative AI Student Declaration

- I have declared all uses of AI tools (e.g., coding assistants, text generators, data analysis tools) in this project.
- I have reviewed and verified all AI-assisted content to ensure accuracy and suitability.
- I take full responsibility for the originality and integrity of my project submission.
- I have referenced all sources, including those suggested or produced with AI support.
- I have not used AI in ways that undermine academic honesty (e.g., plagiarism, unverified code submission).
- My use of AI was supplementary and did not replace my own independent learning and problem-solving.

Signed: Somaan Salim Mirza Date: 18/08/26

Guidance notes for students

- Students are strongly encouraged to review the following resources
    - Student Guide to Gen AI https://www.qmul.ac.uk/library/academic-skills/student-guide-to-generative-ai/
    - Academic Integrity and Misconduct Policy https://www.qmul.ac.uk/governance-and-legal-services/media/dgls-media/policy/current-policies/Academic-Integrity-&-Misconduct-Policy.pdf

- The use of essay mills, homework help sites, plagiarism, collusion, falsification, impersonation, or actions that may give an unfair advantage is strictly prohibited in your final year project. Please refer to the QMUL Academic Integrity and Misconduct policy above.
- It is never acceptable to use GenAI tools to write your entire literature review and submit it as your own work; doing so is considered a form of plagiarism since GenAI outputs are derived from ideas of human authors without proper referencing.
- You may use GenAI tools for checking grammar and spelling, provided the tool is not rewriting a large portion of your work for your final year project.
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- If you are ever unsure whether GenAI is permitted for any part of your project always ask your project supervisor for explicit clarification.
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- Attempting to pass off GenAI-generated content as your own can result in disciplinary investigation under academic misconduct regulations; you may be asked to attend a viva to confirm authorship.
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- Please note that GenAI is helpful for gathering background information, but outputs may be biased, inaccurate, outdated, or fabricated.
- Always consult and cite original, credible sources; GenAI should never be treated as an original source and should not be cited as an author or in the reference list.
- Critically assess all GenAI output for accuracy, double-check any facts, quotations, or citations it suggests especially if you use them in your literature review or other chapters.
- If GenAI has been used for drafting, planning, or structuring, you must clearly acknowledge and document its use in your submission. Therefore, you should keep drafts and records of your own process to show your independent scholarly work and reading notes
- GenAI use may not require acknowledgement for:
    - Familiarising yourself with a topic or research landscape.
    - Summarising information to prepare for more in-depth research.
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- When in doubt, transparency and honesty about your methods and sources including GenAI are expected and protect your academic integrity